

CloudTrail & CloudWatch

1. Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.

Steps:

AWS Console → Search bar → CloudTrail → CloudTrail

Create Trail: Trail name: sunith-monitoring-trail

S3:Enabled

CloudWatch Logs: Enabled

Management events: Read + Write

Multi-region: Enabled

Log file validation: Enabled

Verify S3

AWS Console ↓ S3 ↓ Buckets ↓ sunith-cloudtrail-logs-xxxx

Verify CloudWatch Logs : CloudWatch ↓ Logs ↓ Log groups

Conclusion:

The image consists of two screenshots from the AWS Management Console. The top screenshot shows the 'CloudTrail' 'Edit' page for a trail named 'devops-monitoring-trail'. Under 'Event type', 'Management events' is checked. Under 'API activity', 'Read' and 'Write' are checked. There are 'Cancel' and 'Save changes' buttons at the bottom right. The bottom screenshot shows the 'Amazon S3' 'Buckets' page. A green banner at the top says 'Successfully deleted bucket "sunith.online"'. Below, under 'General purpose buckets', there is a table with one bucket: 'aws-cloudtrail-logs-567487919764-21d75c81' in the 'US East (N. Virginia) us-east-1' region, created on 'September 8, 2026, 17:47:01 (UTC+05:30)'. The table has columns for Name, AWS Region, and Creation date.

CloudTrail Trail Configuration

Events Info
Record API activity for individual resources, or for all current and future resources in AWS account. [Additional charges apply](#)

Event type
Choose the type of events that you want to log.

☒ Management events
Capture management operations performed on your AWS resources.

Management events Info
Management events show information about management operations performed on resources in your AWS account.

☐ No additional charges apply to log management events on this trail because this is your first copy of management events.

API activity
Choose the type of events that you want to log.

☒ Read ☒ Write

☐ Exclude AWS KMS events
☐ Exclude Amazon RDS Data API events

[Cancel](#) [Save changes](#)

Amazon S3 Buckets

[General purpose buckets](#) [All AWS Regions](#) [Directory buckets](#)

General purpose buckets (1) Info [Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

	Name	AWS Region	Creation date
☐	aws-cloudtrail-logs-567487919764-21d75c81	US East (N. Virginia) us-east-1	September 8, 2026, 17:47:01 (UTC+05:30)

CloudTrail & CloudWatch

CloudTrail/

Objects

Properties

Objects (2)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions.

☐	Name	Type	Last modified	Size	Storage
☐	us-east-1/	Folder	-	-	-
☐	us-west-2/	Folder	-	-	-

Amazon S3

Buckets

aws-cloudtrail-logs-567487919764-21d75c81

AWSLogs/

567487919764/

CloudTrail/

us-east-1/

2026/

09/

08/

Copy S3 URI

Objects

Properties

Objects (3)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

< 1 > ⚙

☐	Name	Type	Last modified	Size	Storage class
☐	567487919764-CloudTrail-us-east-1-20260908T1225Z-gdx6CUEH68WdU1H.json.gz	gz	September 8, 2026, 17:52:18 (UTC+05:30)	1.4 KB	Standard
☐	567487919764-CloudTrail-us-east-1-20260908T1225Z-t0LwXVUP7Ubx5b.json.gz	gz	September 8, 2026, 17:52:13 (UTC+05:30)	5.1 KB	Standard
☐	567487919764-CloudTrail-us-east-1-20260908T1225Z-XWmiHrSEwHO1BvkZ.json.gz	gz	September 8, 2026, 17:53:00 (UTC+05:30)	1.5 KB	Standard

Trails

Copy events to Lake

Delete

Create trail

☐	Name	Home region	Multi-region trail	ARN	Insights	Organization trail	S3 bucket	Log file prefix	CloudWatch log group
☒	devops-monitoring-trail	US East (N. Virginia)	Yes	arn:aws:cloudtrail:us-east-1:567487919764:trail/devops-monitoring-trail	Disabled	No	aws-cloudtrail-logs-567487919764-21d75c81	-	arn:aws:logs:us-east-1:567487919764:log-group:aws-logs-567487919764-8700b5c4:

CloudWatch

Log management

aws-cloudtrail-logs-567487919764-8700b5c4

Actions

View in Logs Insights

Start tailing

Search log group

Log group details

Log class

Standard

ARN

arn:aws:logs:us-east-1:567487919764:log-group:aws-cloudtrail-logs-567487919764-8700b5c4:

Creation time

13 minutes ago

Retention

Never expire

Stored bytes

-

Metric filters

0

Subscription filters

0

Contributor Insights rules

-

KMS key ID

-

Deletion protection

Off

Data protection

-

Sensitive data count

-

Custom field indexes

[Configure](#)

Transformer

[Configure](#)

Anomaly detection

[Configure](#)

Bearer token authentication

Off

Syslog ingestion

[Info](#)

[Configure](#)

Log streams

Tags

Data protection

Anomaly detection

Metric filters

Subscription filters

Contributor Insights

Field indexes

Transformer

Log streams (4)

Delete

Create log stream

Search all log streams

CloudTrail & CloudWatch

2. Enable SNS for CloudTrail to send alerts via email.

Steps:

AWS Console ↓ Search ↓ SNS ↓ Simple Notification Service

Step 2 — Check existing topics: SNS ↓ Topics : cloudtrail-alerts

Step 3 — Create SNS topic : Choose:Type: Standard

Create email subscription : Open: sunith-cloudtrail-alerts

Set → Protocol → Email Endpoint → enter the email address

Confirm the email

Connect SNS to CloudTrail : CloudTrail ↓ Trails ↓ sunith-monitoring-trail ↓ Edit

SNS notification delivery → Set: Enabled

Select:Existing SNS topic → Choose: → sunith-cloudtrail-alerts

CloudTrail ↓ CloudWatch Logs ↓ Metric Filter ↓ CloudWatch Alarm ↓ SNS ↓ Email

Conclusion:

The screenshot displays the AWS SNS console interface. At the top, a green notification banner states: "Topic sunith-cloudtrail-alerts created successfully. You can create subscriptions and send messages to them from this topic." Below this, the "sunith-cloudtrail-alerts" topic details are shown, including its name, ARN, and type (Standard). The "Subscriptions" tab is active, showing a table with no subscriptions. Below the console, an email confirmation message from "AWS Notifications <no-reply@sns.amazonaws.com>" is shown. The email is marked as spam and contains a "Report as not spam" button. The message text confirms the subscription to the topic "arn:aws:sns:us-east-1:567487919764:sunith-cloudtrail-alerts" and provides a link to "Confirm subscription". It also includes a note about not replying directly to the email and a link to "sns-opt-out".

CloudTrail & CloudWatch



Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:
arn:aws:sns:us-east-1:567487919764:sunith-cloudtrail-alerts:33def6e1-025c-4318-bcdc-f5b1f6d376e5

If it was not your intention to subscribe, [click here to unsubscribe](#).

Log file SSE-KMS encryption | Info

☐ Enabled

▼ Additional settings

Log file validation | Info

☐ Enabled

Recursive logging | Info

☒ Enabled

SNS notification delivery | Info

☒ Enabled

Create a new SNS topic

☐ New

☒ Existing

SNS topic

arn:aws:sns:us-east-1:567487919764:sunith-cloudtrail-alerts

Use: arn:aws:sns:us-east-1:567487919764:sunith-cloudtrail-alerts

sunith-cloudtrail-alerts

←

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✉️

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1 of 35 < >

AWS Notification Message

Inbox x

👤

AWS Notifications

CloudTrail validation message. -- If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe: <https://sns>.

18:17 (0 minutes ago) ☆

👤

AWS Notifications <no-reply@sns.amazonaws.com>

to me ▾

18:17 (0 minutes ago) ☆ 🗑️ ↶ ⋮

```
[{"s3Bucket":"aws-cloudtrail-logs-567487919764-21d75c81","s3ObjectKey":["AWSLogs/567487919764/CloudTrail/ca-central-1/2026/09/08/567487919764_CloudTrail_ca-central-1_20260908T1250Z_EWQvp4kjYgryhrlB.json.gz"]}]
```

...

👤

AWS Notifications <no-reply@sns.amazonaws.com>

to me ▾

18:17 (0 minutes ago) ☆ 🗑️ ↶ ⋮

```
[{"s3Bucket":"aws-cloudtrail-logs-567487919764-21d75c81","s3ObjectKey":["AWSLogs/567487919764/CloudTrail/ap-south-1/2026/09/08/567487919764_CloudTrail_ap-south-1_20260908T1245Z_id5OuXby55qDFw4.json.gz"]}]
```

...

👤

AWS Notifications <no-reply@sns.amazonaws.com>

to me ▾

18:17 (0 minutes ago) ☆ 🗑️ ↶ ⋮

```
[{"s3Bucket":"aws-cloudtrail-logs-567487919764-21d75c81","s3ObjectKey":["AWSLogs/567487919764/CloudTrail/eu-north-1/2026/09/08/567487919764_CloudTrail_eu-north-1_20260908T1250Z_whyUXAhSQvlp7vC.json.gz"]}]
```

...

CloudTrail & CloudWatch

3. Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.

Steps:

AWS Console ↓ CloudWatch

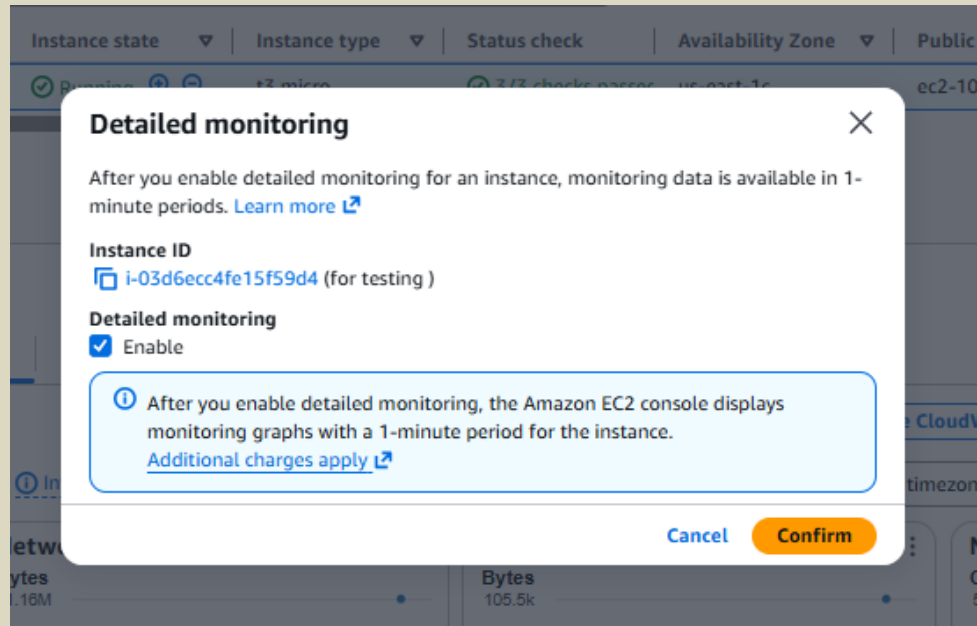
Open Metrics :

CloudWatch ↓ Metrics ↓ All metrics

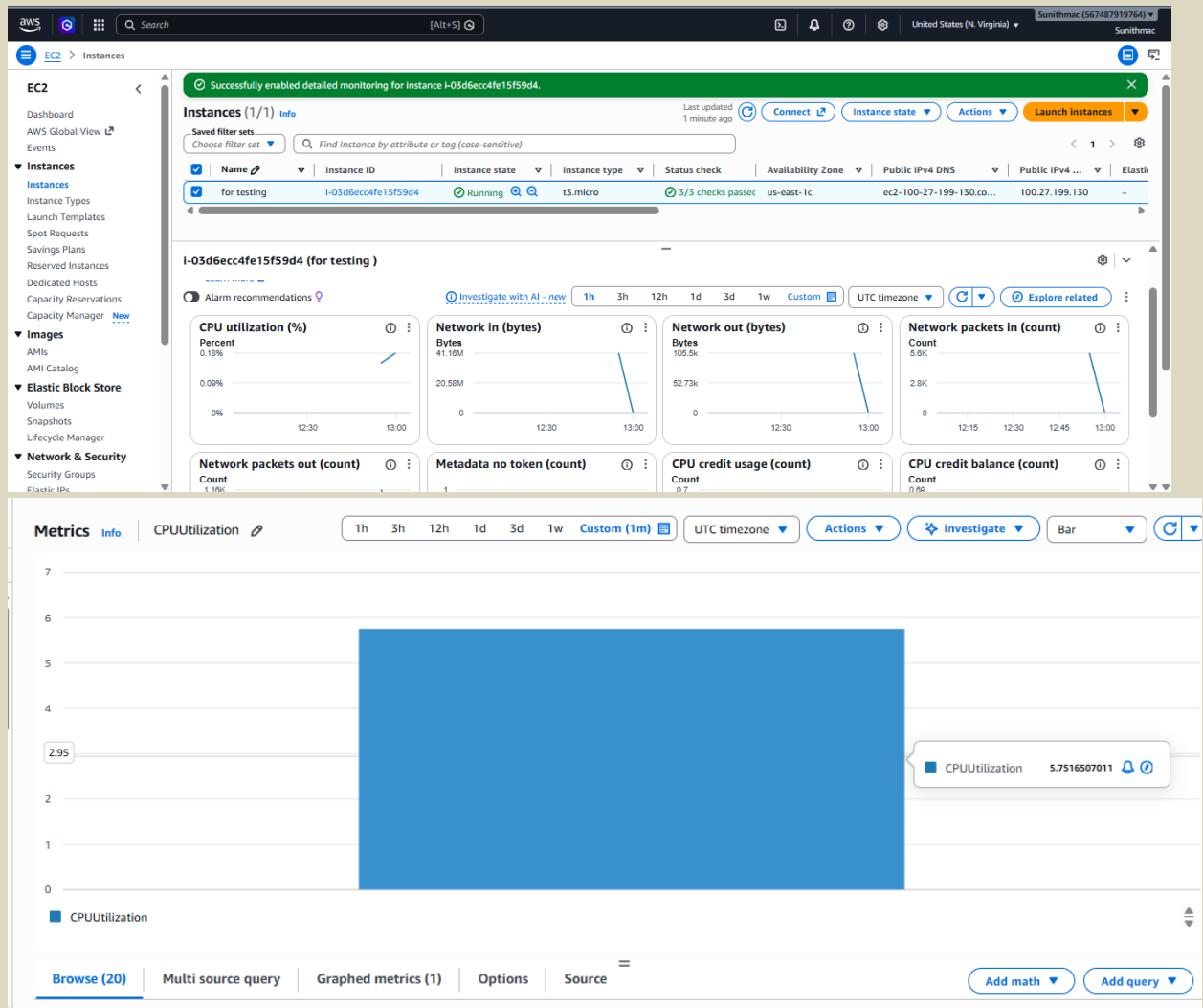
Select: → EC2 → Then: Per-Instance Metrics

Select CPU Utilization

Conclusion:



CloudTrail & CloudWatch



4. Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.

Steps:

Open Alarms

CloudWatch ↓ Alarms ↓ All alarms

Select metric → EC2 ↓ Per-Instance Metrics → Select: → CPUUtilization

Configure statistic → Statistic: Average → set Period: 5 minutes

Configure threshold → Conditions → Threshold type: Static → CPUUtilization Greater than 70

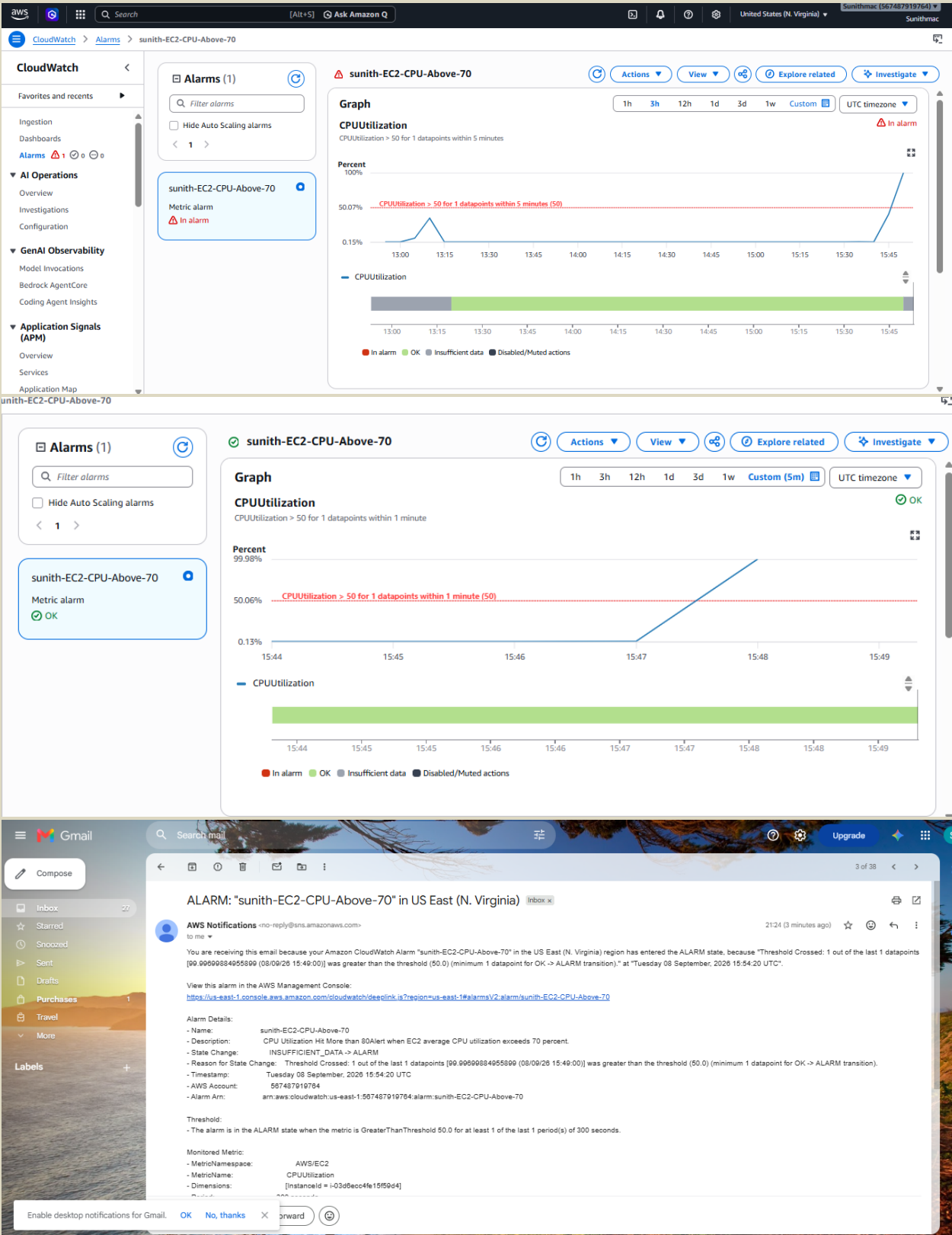
Evaluation period 1 →

Configure SNS → Under: Notification

Choose: In alarm → sunith-cloudtrail-alerts → Alarm name → Next → Create alarm

CloudTrail & CloudWatch

Conclusion:



CloudTrail & CloudWatch

5. Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert.

Steps:

Downloaded tomcat to linux system

wget <https://d1cdn.apache.org/tomcat/tomcat-9/v9.0.121/bin/apache-tomcat-9.0.121.tar.gz>

extracted → tar -xvf apache-tomcat-9.0.121.tar.gz

installed java -17

cd apache-tomcat-9.0.121 → cd bin/ → ./startup

created system service file for tomcat

and configured with Security Credentials

added script to

```
#!/bin/bash
```

```
#set -x
```

```
TOKEN=$(curl -sX PUT "http://169.254.169.254/latest/api/token" \
-H "X-aws-ec2-metadata-token-ttl-seconds: 21600")
```

```
INSTANCE_ID=$(curl -s \
-H "X-aws-ec2-metadata-token: $TOKEN" \
http://169.254.169.254/latest/meta-data/instance-id)
```

```
checkTomcatStatus(){
    counter=0
    ps x | grep /opt/apache-tomcat-9.0.121/ | grep -v grep |
    while read -r LINE
    Do
        read PROCESS_ID <<< $LINE
        counter=$((counter+1))
        echo $counter
    Done
}
```

```
i=$(checkTomcatStatus)
```

```
if [ "$i" == "" ]
```

```
then
```

```
    aws --region us-east-1 cloudwatch put-metric-data --metric-name tomcat --value 0 --
    namespace tomcat --dimensions InstanceId=$INSTANCE_ID
```

```
Else
```

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```
aws --region us-east-1 cloudwatch put-metric-data --metric-name tomcat --value 1 --  
namespace tomcat --dimensions InstanceId=$INSTANCE_ID  
Fi  
# Need to change the region
```

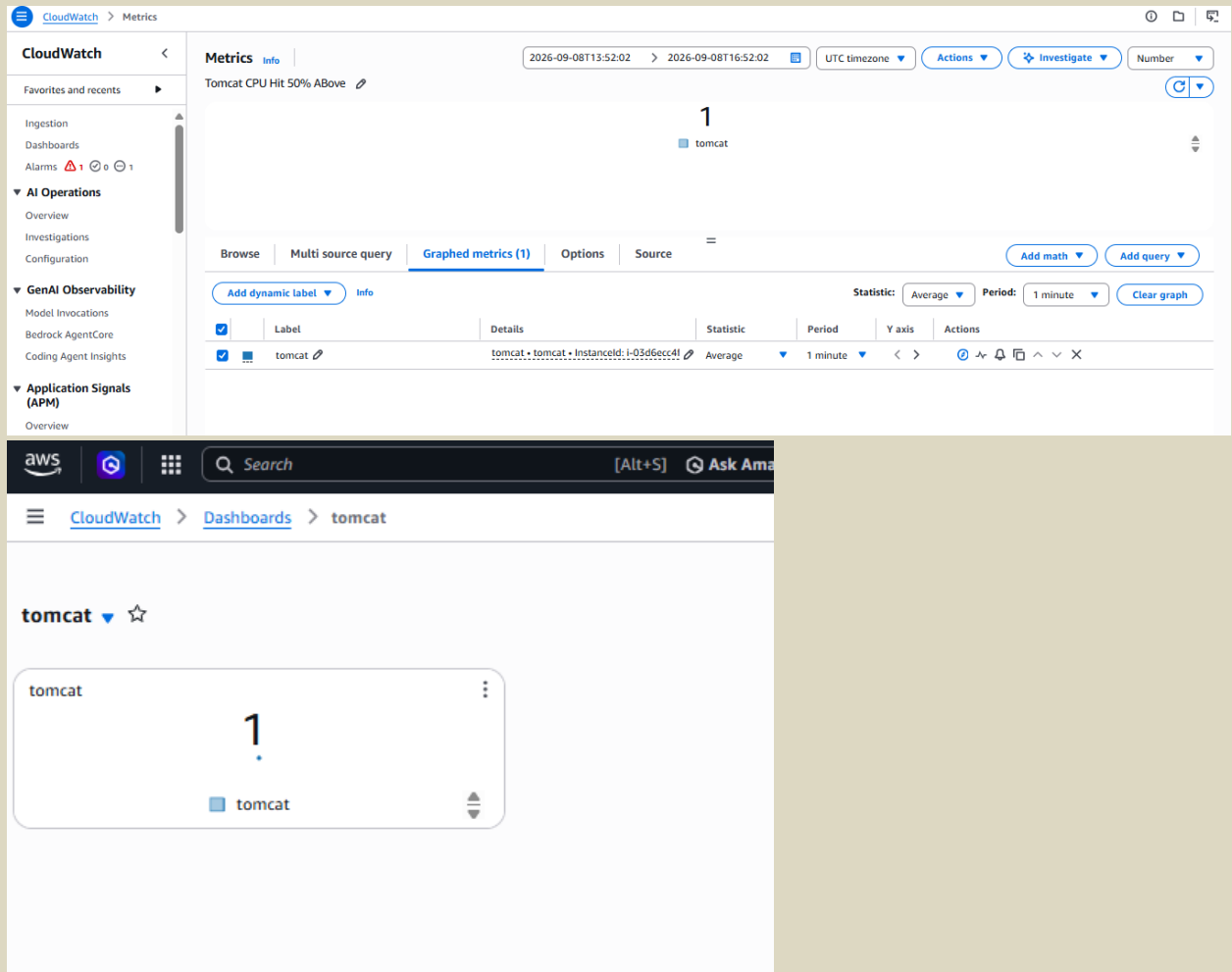
Conclusion:

```
[root@ip-172-31-16-69 ~]# cat /etc/hosts
172.31.16.69 ip-172-31-16-69.ec2-user

INSTANCE_ID=$(curl -s \
-H "X-aws-ec2-metadata-token: $TOKEN" \
http://169.254.169.254/latest/meta-data/instance-id)

echo "$INSTANCE_ID"
i-03d6ecc4fe15f59d4
[root@ip-172-31-16-69 ec2-user]# Invalid length for parameter MetricData[0].Dimensions[0].Value,
value: 0, valid min length: 1
bash: Invalid: command not found
bash: value:: command not found
[root@ip-172-31-16-69 ec2-user]# cat -n monitoring.bash
 1  #!/bin/bash
 2
 3  #set -x
 4  INSTANCE_ID=$(/opt/aws/bin/ec2-metadata -i | cut -d " " -f2)
 5
 6  checkTomcatStatus(){
 7      counter=0
 8      ps x | grep /opt/apache-tomcat-9.0.121/ | grep -v grep | #Need to change tomcat version
 9      while read -r LINE
10      do
11          read PROCESS_ID <<< $LINE
12          counter=$((counter+1))
13          echo $counter
14      done
15  }
16
17  i=$(checkTomcatStatus)
18  #echo $i
19
20  if [ "$i" == "" ]
21  then
22      aws --region us-east-1 cloudwatch put-metric-data --metric-name tomcat --value 0 --namespace tomcat --dimensions InstanceId=$INSTANCE_ID
23  else
24      aws --region us-east-1 cloudwatch put-metric-data --metric-name tomcat --value 1 --namespace tomcat --dimensions InstanceId=$INSTANCE_ID
25  fi
26
27  # Need to change the region
28
[root@ip-172-31-16-69 ec2-user]# vi monitoring.bash
[root@ip-172-31-16-69 ec2-user]# ls
monitoring.bash
```

CloudTrail & CloudWatch



6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

CloudWatch Dashboard – Nginx Monitoring

Aim: Create a CloudWatch dashboard to monitor Nginx and send an alert when Nginx is stopped.

Steps

Verify IAM Role

```
aws sts get-caller-identity
```

Create Nginx Check Script

```
sudo vi /usr/local/bin/check_nginx.sh
#!/bin/bash
if systemctl is-active --quiet nginx; then
    echo 1
Else
    echo 0
Fi
sudo chmod +x /usr/local/bin/check_nginx.sh
```

CloudTrail & CloudWatch

Create CloudWatch Metric

```
aws cloudwatch put-metric-data \  
--namespace Nginx \  
--metric-name NginxRunning \  
  
--value $(/usr/local/bin/check_nginx.sh)
```

Create Dashboard

AWS Console → **CloudWatch** → **Dashboards** → **Create Dashboard**

Select:

Nginx → NginxRunning

If the dashboard shows --, wait a few minutes and verify that the metric is being published and that the AWS Region is correct.

Create Alarm

CloudWatch → **Alarms** → **Create Alarm**

Condition:

NginxRunning < 1

Configure **SNS** → **Email notification**.

Test

Stop Nginx:

```
sudo systemctl stop nginx
```

The alarm should change to **ALARM** and send an email.

Start it again:

```
sudo systemctl start nginx
```

The alarm should return to **OK**.

Result

Nginx is monitored using CloudWatch, and an email alert is generated when the service is not running.

CloudTrail & CloudWatch

